

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

- **HTTP (Hypertext Transfer Protocol):** An application-layer protocol used primarily for transmitting web page data between a browser and a server.
- **HTTPS (Hypertext Transfer Protocol Secure):** An extension of HTTP that uses SSL/TLS encryption to provide secure, authenticated communication over a computer network.
- **FTP (File Transfer Protocol):** A standard network protocol used for the transfer of computer files between a client and server on a computer network.
- **SMTP (Simple Mail Transfer Protocol):** A communication protocol specifically designed for sending and routing electronic mail between mail servers.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4 uses a 32-bit addressing scheme providing approximately 4.3 billion unique addresses, typically represented in decimal format (e.g., 192.168.1.1). In contrast, IPv6 uses a 128-bit addressing scheme, offering a virtually inexhaustible number of addresses represented in hexadecimal format (e.g., 2001:db8::ff00:42:8329).

3. Run nslookup google.com on your system and write down the IP address you get.

The IP Address: **192.168.29.1**

4.What do these HTTP status codes mean: 200, 301, 404, 500, 403?

HTTP response status codes indicate whether a specific HTTP request has been successfully completed.

Status Code	Meaning	Category
200	OK	Success: The request has succeeded.
301	Moved Permanently	Redirection: The requested resource has been assigned a new permanent URI.
403	Forbidden	Client Error: The server understood the request but refuses to authorize it.
404	Not Found	Client Error: The server cannot find the requested resource.
500	Internal Server Error	Server Error: The server encountered an unexpected condition.

5.Explain DNS hierarchy (root, TLD, domain, subdomain) with **www.google.com** as an example.

Using **www.google.com** as an example, the Domain Name System (DNS) follows a distributed hierarchical structure:

- **Root Level:** Represented by an implicit dot at the end of the domain (e.g., `www.google.com.`), managed by root name servers.

- **Top-Level Domain (TLD):** The final segment of the domain name, which in this case is **.com**.
- **Domain (Second-Level):** The specific name registered to an entity, which in this case is **google**.
- **Subdomain:** A prefix added to the domain to categorize a specific section or service, which in this case is **www**.